Roll No.............

II BE CLASS TEST-II OCTOBER 2023

Computer Engineering

3ACRC1 Applied Mathematics-III

Time-70 mins. Maximum Marks [20]

Note-Attempt all the questions.

Q.1. Perform 8 iterations of the relaxation method to solve the system of equations- [4]

$$\begin{aligned} 1.5x - 7y + 5z &= -2 \\ 5x + 2.7y - 4z &= 7 \\ -3x + 9y - 1.8z &= -3 \end{aligned}$$

Q.2. Use $R-K$ method of fourth order to find $y(2.2)$ in two steps from- [4]

$$\frac{dy}{dx} = -\frac{ye^{xy}}{xe^{xy} + 2y}; y(1.5) = 2$$

Q.3. The amount A of a substance remaining in a reacting system after an interval of time t in a certain chemical experiment is tabulated below. Obtain the rate of change of the amount remaining after time $t = 9$ min. [4]

t(min)	2	5	8	11	14	17
A(gm)	94.8	87.9	81.3	75.1	72.3	67.8

Q.4. Use Simpson's 1/3 rule to compute the integral taking 11 ordinates- [4]

$$\int_{1/2}^{\pi/2} \frac{\cos\sqrt{x} + \ln x^2}{\sqrt{\log(1+x) + e^{-x}}} dx$$

Q.5. Using Lagrange's interpolation formula, find the pull P required to lift a load $W = 105$ kg by means of a pully block from the following table- [4]

W	50	70	110	130	160
P	12	15	21	23	26

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